THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET B

Chapter 4: **Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. When current flows in a wire near a compass needle, the needle: 1

(a) stays unaffected **(b)** melts
(c) stops the current **(d)** gets deflected

2. The strength of an electromagnet increases when: 1

(a) more cells are connected **(b)** the circuit is opened
(c) the iron core is removed **(d)** fewer cells are used

3. Which material is attracted by an electromagnet? 1

(a) plastic **(b)** glass **(c)** iron **(d)** wood

4. Heating effect of current is due to the: 1

(a) colour of the wire **(b)** resistance of the conductor
(c) magnetism of the wire **(d)** length of the cell

5. Which appliance is based on the heating effect of current? 1

(a) electric iron **(b)** torch bulb only
(c) compass **(d)** electromagnet crane

6. A voltaic cell has two different metal rods and a: 1

(a) insulator **(b)** magnet **(c)** electrolyte **(d)** bulb

7. Who built the first battery using metals and a wet medium? 1

(a) Oersted **(b)** Luigi Galvani **(c)** Alessandro Volta **(d)** Edison

8. Which pair of metals can make a voltaic cell? 1

(a) iron and iron **(b)** zinc and copper
(c) zinc and zinc **(d)** copper and copper

9. A dry cell is: 1

(a) a magnet **(b)** a switch
(c) rechargeable **(d)** a single-use cell

10. The magnetic field around a current-carrying wire can be shown using a: 1

(a) spring balance **(b)** ammeter only **(c)** thermometer **(d)** compass

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State one use of a lifting electromagnet. 2

12. What is the heating effect of electric current? 2

13. Why is nichrome used in heating elements? 2

14. Name two appliances that use heating element. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe three applications of the heating effect of electric current. 3

16. Why should plugs and wires of proper rating be used? 3

17. Explain the working principle of a voltaic cell. 3

OR

Describe how Galvani's and Volta's observations led to the first battery. 3

18. Describe the lemon cell activity. 3

OR

Name three metal pairs that can be used in voltaic cells and state which usually acts as the positive electrode. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Describe the structure and working of a dry cell with a labelled diagram. 5

OR

Discuss the story of Galvani and Volta and its importance in the development of the battery. 5

20. Explain how the lemon cell activity is done and what it teaches about voltaic cells. 5

OR

Compare dry cells and rechargeable batteries on use, life, cost and environment. Explain e-waste. 5

END OF QUESTION PAPER — Total: 40 Marks

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